Agentic Credit Risk Intelligence Engine

Milestone 2 Project Report

Retrieval-Augmented Generation, LangGraph Orchestration, and LLM-Powered Risk Reporting

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1. Abstract

Phase 2 of the Intelligent Credit Risk Scoring Engine transforms a static ML classifier into a fully agentic AI system. A three-node LangGraph pipeline connects a **LightGBM + SMOTE** risk classifier (80.41% accuracy, Macro F1 0.71, 100 features), a ChromaDB RAG module with **2,931 regulatory chunks** from three official RBI/SEBI documents, and a Groq-hosted `llama-3.3-70b-versatile` model that synthesizes all signals into a structured, grounded JSON risk report. The full pipeline — from raw bureau input to explainable credit assessment — completes in approximately **10 seconds**.

2. Introduction

Credit risk classification answers *how risky* a borrower is. Phase 1 answered that well — a behavioral ML classifier trained on CIBIL bureau data predicted risk tiers (P1–P4) at 78% accuracy, deliberately excluding `Credit_Score` to learn from raw signals like enquiry frequency, delinquency history, and payment patterns.

Phase 2 answers the harder questions: *Why? What should the bank do? What does regulation say?* It wraps the Phase 1 model in an **agentic pipeline** that retrieves relevant lending guidelines and produces a document-backed advisory report for credit officers — not just a label.

3. Phase 2 Architecture — The Agentic Layer

Three sequential LangGraph nodes share a typed state object (`TypedDict`). Each node has a single responsibility. If a node fails it sets an error flag; downstream nodes detect it and skip, preventing silent failures. The shared state carries: `prospect_id`, raw `features`, the assigned `risk_tier`, `probabilities`, `top_risk_factors`, `retrieved_chunks`, the final `report`, and an `error` field.

LangGraph Workflow — End-to-End Flow

Raw Bureau Features (Streamlit UI)

↓

Node 1 — Risk Analyzer

LightGBM + SMOTE · Risk tier, class probabilities, top 5 features

↓

Node 2 — Guideline Retriever

ChromaDB + all-MiniLM-L6-v2 · 4 most relevant regulatory chunks

↓

Node 3 — Report Generator

Groq llama-3.3-70b-versatile · Structured JSON risk assessment

↓

Final Report displayed in Streamlit UI

4. Node 1 — Risk Analyzer

4.1 Preprocessing Pipeline

The preprocessing exactly mirrors the training pipeline to eliminate train-serve skew. In order: sentinel values (-99999) are replaced with `NaN`; leakage columns (`Credit_Score`, `CC_utilization`, `PL_utilization`) are dropped; delinquency nulls are filled with 0; five composite behavioral features are computed; categorical variables are one-hot encoded (`drop_first=True`); and the result is reindexed to exactly the 100 training columns, with any missing column filled with 0.

4.2 Engineered Features

Feature	Formula	What It Captures
`delinq_burden`	$D_{\text{count}} \times D_{\text{level}}$	Severity-weighted delinquency load
`enq_pressure`	$E_{3m}/(E_{12m}+1)$	Recency ratio of credit enquiries
`account_health`	$N_{\text{std}}/(N_{\text{sub}}+N_{\text{dbt}}+N_{\text{lss}}+1)$	Standard vs. distressed account ratio
`income_stability`	$\text{Income} \times \ln(1+T_{\text{employ}})$	Income weighted by employment tenure
`recency_risk`	$1/(T_{\text{deliq}}+1)$	Inverse decay of delinquency recency

Table 1: Engineered behavioral features computed at inference (identical to training pipeline).

4.3 Model Inference

The frozen LightGBM model (`models/finalized_model.joblib`) is invoked via the low-level `Booster.predict()` API. Three values are written to shared state: the risk tier (P1–P4 from `argmax`), class probabilities for all four tiers, and the top 5 features by importance with their live input values.

5. Node 2 — Guideline Retriever (RAG)

The Guideline Retriever builds a natural-language query from the assigned risk tier and top three risk factors, embeds it locally using `all-MiniLM-L6-v2` (384-dim, no external API call), and searches the ChromaDB vector store via cosine similarity. The **4 most relevant chunks** are returned with their source document names and distance scores, then passed directly into the Node 3 prompt.

6. Node 3 — Report Generator

6.1 Prompt Design

The system prompt constrains the LLM as a bank credit risk analyst with strict grounding rules: use only facts from the retrieved guidelines, respond with valid JSON only, and state "Insufficient guideline context" when guidelines do not cover a point. The user message includes the prospect ID, risk tier, model probabilities, top risk factors, and the four retrieved chunks (`temperature=0.1`, `max_tokens=1200`).

6.2 Output JSON Schema

Section	Field	Description
`summary`	`risk_tier`	Assigned tier: P1, P2, P3, or P4
	`severity`	Human label: Very Low / Low / Medium / High
	`recommendation`	One-sentence lending action for the credit officer
	`overview`	2–3 sentence narrative of the borrower's risk profile
`deep_dive`	`top_risk_factors`	Top 3 features driving the risk tier
	`guideline_references`	Direct quotes or paraphrases from retrieved regulations
	`suggested_actions`	Three concrete actions recommended for the credit team
`confidence`	`P1-P4`	Class probabilities from LightGBM (not the LLM)

Table 2: Structured JSON schema enforced on Node 3 output.

After parsing the LLM response, the `confidence` block is **overwritten** with the ML model's actual probabilities. The LLM's self-reported certainty is always discarded — probability scores must come from the calibrated LightGBM model.

7. Knowledge Base Construction

7.1 Source Documents

The RAG knowledge base is built offline by `ingest.py` and stored persistently in `chroma_db/`. Three official regulatory documents totalling **514 pages** were selected:

Document	Source	Pages
RBI Master Circular — Basel III Capital Regulations	RBI	328
IRACP Norms — Income Recognition & Asset Classification	RBI	77
SEBI Master Circular — Credit Rating Agencies	SEBI	109
Total		**514**

Table 3: Regulatory documents in the RAG knowledge base.

7.2 Chunking and Embedding

Documents are split with a 500-character sliding window (100-character overlap, minimum 50 characters) producing **2,931 chunks**. All chunks are embedded offline in a single batch pass using `all-MiniLM-L6-v2` (`batch_size=64`) and stored in ChromaDB with a cosine-distance HNSW index. At query time only the search query is re-embedded; document vectors are retrieved from disk.

8. LightGBM Upgrade and Model Comparison

Phase 1 used HistGradientBoosting with cost-sensitive weighting. Phase 2 upgrades to **LightGBM + SMOTE** to address severe class imbalance (P2 alone is 62.7% of records). SMOTE synthetically generates minority-class samples before training, producing stronger decision boundaries for P1, P3, and P4.

Model	Approach	Accuracy	Macro F1	P3 Recall
Decision Tree	Baseline (`max_depth=5`)	77.4%	0.67	0.30
Random Forest	GridSearchCV tuning	78.4%	0.71	0.41
LightGBM + SMOTE	**SMOTE + gradient boosting**	**80.4%**	**0.71**	**0.32**

Table 4: Model comparison across all experiments. LightGBM + SMOTE is the final deployed model.

9. Tools and Technologies

Category	Tool / Library	Role
ML & Balancing	LightGBM, imbalanced-learn	Risk classification with SMOTE oversampling
Orchestration	LangGraph	Stateful three-node agentic workflow
LLM	Groq / llama-3.3-70b	Structured JSON report generation
Embeddings	all-MiniLM-L6-v2	384-dim semantic query vectors
Vector Store	ChromaDB	Regulatory chunk storage and cosine retrieval
UI	Streamlit	Interactive web interface
Data	pandas, NumPy, pypdf	Feature engineering and PDF ingestion

Table 5: Core tools and technologies used in Phase 2.

10. Results

11. Conclusion

Phase 2 delivers a complete agentic credit risk intelligence system. The **LightGBM + SMOTE** classifier achieves 80.4% accuracy and drives all downstream reasoning. The **RAG pipeline** grounds every recommendation in real regulatory content from RBI and SEBI circulars, preventing hallucination. The **LangGraph orchestration** ties everything together in a fault-tolerant, explainable workflow that a credit officer can act on — not just a risk label, but a full advisory

Category	Metric	Value
Model Performance	Accuracy (LightGBM + SMOTE)	**80.41%**
	Macro F1-Score	**0.71**
	Test Set / Features	10,268 samples / 100 features
Pipeline Timing	Node 1 — LightGBM inference	<1 s
	Node 2 — ChromaDB retrieval	1–3 s
	Node 3 — Groq LLM report	4–7 s
Knowledge Base	Chunks / Documents	2,931 chunks / 3 regulatory docs (514 pages)

Table 6: Phase 2 system results — model performance, pipeline timing, and knowledge base.

report with factors, guidelines, and actions, delivered in under 10 seconds.

Report prepared by Team PowerPuff Boys — Agentic Credit Risk Classification, April 2026

LightGBM + SMOTE · LangGraph · ChromaDB · Groq llama-3.3-70b-versatile · Streamlit